Performance Analysis of an IR-Based Leader-Follower System on Circular Trajectories

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Abstract—**Navigating curved paths is a significant challenge for leader-follower robot systems, especially when using low-cost infrared (IR) sensors. This paper investigates these stability challenges using a system of two Pololu 3pi+ robots. The leader creates a path by projecting an IR trail while driving in circles, and the follower attempts to maintain the link. We conducted a series of experiments to test how the follower performs on three different circle sizes. By systematically changing the Base Speed and Proportional Gain (K_p), we identified the limits of the system. The results show that tighter circles are much harder to track, requiring a higher Gain setting (K_p) to react quickly enough. However, increasing this responsiveness makes the robot unstable at high speeds. This study demonstrates a clear trade-off: to maintain stability on sharp curves, robot speed must be constrained.**

I. Introduction

Multi-robot coordination is a fundamental requirement for autonomous systems in logistics and search-and-rescue applications [5]. Among formation strategies, the leader-follower architecture reduces system complexity by centralising trajectory planning in a single agent, allowing secondary agents to track the leader using local sensor data.

This project implements a low-latency, optical leader-follower system using two Pololu 3pi+ 32U4 mobile robots [4]. Unlike radio-frequency methods (e.g., Bluetooth or Wi-Fi), which can suffer from bandwidth saturation or signal interference, this system utilizes implicit communication via Infrared (IR) intensity. The primary engineering challenge investigated in this report is the stability of this optical link when navigating variable curvature paths.

A. System Topology and Sensing Mechanism

To interpret the experimental results, it is necessary to define the hardware constraints of the Pololu 3pi+ platform. The system relies on a specific emitter-receiver configuration:

- **The Leader (Emitter):** The leader robot is kinematically inverted (programmed to navigate in reverse). This orientation positions the robot's front-facing IR proximity sensors towards the follower. By driving the shared emitter control pin (Pin 11) to a logical `LOW` state, the robot projects a continuous-wave IR signal onto the navigation surface, creating a dynamic "virtual rail."
- **The Follower (Receiver):** The follower utilizes a downward-facing 5-channel reflectance sensor array. Specifically, the system samples the analog voltage from the inner-left (DN2, Pin A0) and inner-right (DN4, Pin A3) phototransistors.

Sensor Physics: In this configuration, the sensors do not detect reflectance properties (as in traditional line following). Instead, they measure the intensity of the IR light reflected from the floor surface. The phototransistors operate effectively as a differential pair; a difference in intensity between DN2 and DN4 indicates the follower's lateral deviation from the leader's trajectory center.

While IR sensing provides a robust, low-cost solution for indoor environments, it introduces a non-linear control challenge. The signal intensity decays rapidly with distance (following the Inverse Square Law), and the target is dynamic. Consequently, the controller must balance **responsiveness** (tracking sharp turns) against **stability** (avoiding oscillation) [3].

B. Hypothesis Statement

Based on the physical limitations of the IR sensors and the differential drive motors, we propose that the robot's success depends on three specific factors.

1) **Optimal Gain Range:** The Proportional Gain (K_p) will have a specific "safe zone." If K_p is too low (near 0), the robot will react too slowly and drift off the path (*Under-steering*). Conversely, if K_p is too high (near 2.0), the robot will over-react to small errors and start shaking (*Oscillation*). We predict the optimal value will be a central balance (approx. 0.8) that provides enough turning power without causing instability.
2) **Velocity-Curvature Link:** There is no single "best speed" for all situations. We predict that the optimal Base Speed is strictly dependent on the circle diameter. A speed that works well for a large circle may fail on a tight circle due to the sharp change in angle.
3) **Environmental Sensitivity:** The physical environment is a critical variable. We hypothesize that changes in surface texture will drastically alter the friction and sensor readings. A control setting that works perfectly on one floor type is expected to fail on a different surface, indicating that the system is highly sensitive to environmental conditions.

C. Hypothesis Statement

Based on the limited field-of-view of the reflective sensors and the non-linear dynamics of the differential drive system, we propose the following specific hypotheses regarding the system's stability:

1. The Stability-Curvature Trade-off: We hypothesize that there is an inverse relationship between **path curvature** and **maximum stable velocity**. As the leader's path becomes tighter (smaller radius), the follower requires a higher Proportional Gain (K_p) to compensate for the rapid angular displacement. However, increasing K_p reduces the system's damping against noise. Therefore, we predict that successful navigation of tight trajectories will strictly require a reduction in Base Speed to prevent oscillatory instability.

2. The Optimal Gain Range: We predict that the Proportional Gain (K_p) must operate within a specific, quantifiable window for effective tracking.

- **Lower Bound:** If K_p is below this threshold, the system will succumb to **under-steering**, failing to react fast enough to turn, resulting in signal loss.
- **Upper Bound:** If K_p exceeds this threshold, the system will exhibit **over-steering**, leading to sustained oscillation ("wobble") and eventual divergence from the path.

3. Environmental Sensitivity: We hypothesize that the system is highly sensitive to surface characteristics. We predict that a change in floor texture will significantly alter the wheel friction and IR reflectance properties, causing a control configuration that is stable on one surface to fail on another without parameter re-tuning.

Report Structure: The rest of this report proceeds as follows: Section II describes the implementation of the robotic system and control logic. Section III details the experimental methodology used to test these hypotheses. Section IV presents the results of the parameter sweep, and Section **??** discusses the implications of these findings before concluding the report..

II. Implementation

This section details the software architecture and control logic developed for the Pololu 3pi+ robots. The system was designed to operate autonomously using a closed-loop feedback mechanism illustrated in the system flowchart (Fig. 2).

A. Preliminary Sensor Analysis

Before implementing the controller, the response characteristics of the IR configuration were characterized. A static test was conducted by manually varying the distance between the leader (emitter) and follower (receiver) to map the sensor intensity.

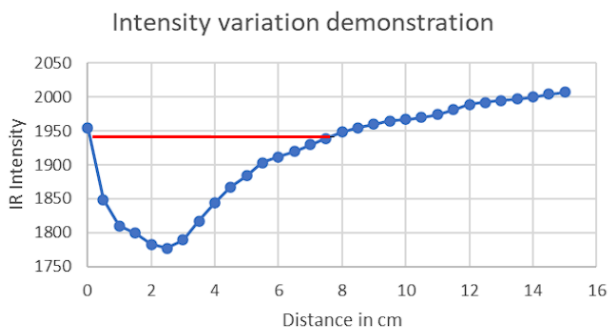

Fig. 1. IR Intensity Curve demonstrating the non-linear response of the reflectance sensors to the leader's projected signal. The red line highlights the effective linear tracking region.

As shown in Fig. 3, the relationship between distance and sensor intensity is non-linear. The intensity reaches a local minimum (peak signal strength) at close proximity ($\approx$ 2.5cm). As the distance increases beyond 3cm, the intensity value rises (indicating lower light) following an inverse-square relationship.

- **Significance:** This phenomenon occurs due to the horizontal propagation of the IR emitter signals from the front end of the leader robot relative to the downward-facing receiver. The "Area Under Curve" (highlighted in red) represents the stable linear region where tracking is viable. The control logic was specifically tuned to operate within this region.

B. Control Architecture

The core of the follower's behavior is a Proportional (P) Controller designed to minimize lateral deviation.

1) Error Formulation: The deviation from the path center, denoted as error $e(t)$, is calculated as the differential between the inner-left (S_L) and inner-right (S_R) sensor readings:

$$e(t) = S_L(t) - S_R(t) \tag{1}$$

A positive error indicates the leader is to the left, requiring a left turn, while a negative error indicates the opposite.

2) Proportional Control Law: To correct this error, a steering adjustment $u(t)$ is computed by multiplying the error by a tunable Proportional Gain (K_p):

$$u(t) = K_p \cdot e(t) \tag{2}$$

A larger error produces a stronger correction. A well-tuned K_p minimizes oscillation while ensuring responsive steering. This adjustment is applied differentially to the robot's base velocity (V_{base}):

$$V_L = V_{base} + u(t) \tag{3}$$

$$V_R = V_{base} - u(t) \tag{4}$$

3) Safety and Deceleration Logic: To prevent collision or divergence, a deceleration state is triggered if the aggregate sensor reading ($S_{total} = S_L + S_R$) exceeds the effective tracking threshold. In this state, velocity is reduced by a fixed decrement (δ) per cycle:

$$V_{new} = V_{current} - \delta \tag{5}$$

C. Algorithm Design

The complete logic flow is summarized in Algorithm 1 and the accompanying flowchart (Fig. 2).

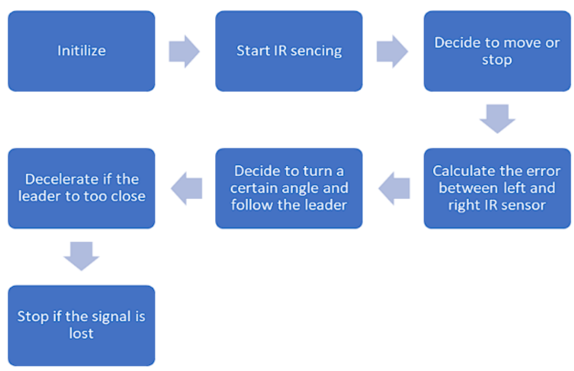

Fig. 2. Control logic flowchart describing the decision-making process for the follower robot.

Algorithm 1 Leader Tracking Logic

```
Initialize Motors and Sensors
loop
    S_L ← ReadLeftSensor()
    S_R ← ReadRightSensor()
    S_total ← S_L + S_R
    if S_total > Threshold then
        Call Decelerate(δ)                ▷ Signal Lost/Too Far
    else
        e(t) ← S_L − S_R
        u(t) ← e(t) × K_p
        V_L ← V_base + u(t)
        V_R ← V_base − u(t)
        SetMotors(V_L, V_R)
    end if
end loop
```

III. Experiment Methodology

The primary objective of this study was to determine the stability boundaries of the leader-follower system under different curvature conditions. Given the absence of active collision avoidance, the experiment focused on identifying the optimal control parameters that allow the follower to maintain path fidelity across varying path curvatures without divergence.

A. Overview of Method

An iterative parameter sweep method was employed. The experiment was structured into three distinct phases corresponding to the three specific circular paths generated by the leader:

1) **Small Curvature:** Diameter $D \approx 20.5$ cm
2) **Medium Curvature:** Diameter $D \approx 28.0$ cm
3) **Wide Curvature:** Diameter $D \approx 40.0$ cm

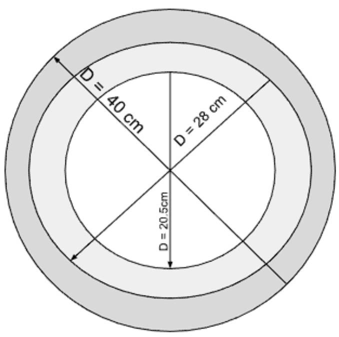

Fig. 3. Schematic representation of the three experimental path geometries. The distinct diameters ($D = 20.5$ cm, $D = 28$ cm, and $D = 40$ cm) correspond to the High, Medium, and Low curvature test cases used to evaluate the system's tracking stability.

For each curvature, the system was subjected to a "Stress Test" protocol. We systematically varied the Proportional Gain (K_p) and Base Speed to observe the system's response.

Reliability Protocol: To ensure statistical reliability and mitigate variance caused by sensor noise, the primary optimization phase on the smooth surface involved $N \geq 20$ **repeated trials** for each optimal configuration.

B. Discussion of Variables

1) Independent Variables:

- **Proportional Gain (K_p):** This variable determines the steering "stiffness." It was varied experimentally to find the balance point between under-steering (drifting out) and over-steering (oscillation).
- **Base Speed:** The forward velocity was varied (typically within the range [25, 30]) to test the dynamic limits of the sensors.
- **Path Curvature:** The three discrete diameters defined above (20.5 cm, 28 cm, 40 cm).
- **Surface Texture:** A secondary comparative study was conducted between a **Smooth Surface** (Standard Laminate) and a **Wooden Surface** to evaluate environmental sensitivity.

2) Dependent Variables:

- **Path Completion Status:** A binary record of whether the robot maintained the "virtual link" for the full duration of the trial.
- **Qualitative Stability:** Visual observation of the robot's behavior (e.g., smooth tracking vs. oscillatory motion).

3) Controlled Variables:

- **Environment:** A constant indoor lighting environment was maintained to prevent IR interference.
- **Start Position:** For every trial, the follower was manually placed such that the **sensor-to-sensor distance was exactly 3 cm**. This ensures the system initializes at the peak intensity point (refer to Sensor Analysis).

C. Discussion of Metric

Given the high number of repetitions ($N > 20$), a quantitative performance metric was derived to measure reliability.

Success Rate (S_{rate}) A single trial is recorded as a binary outcome (r_i), where $r_i = 1$ if the follower maintains the link for the entire trajectory, and $r_i = 0$ if it triggers the safety stop. The final metric is the percentage of successful runs over the total number of trials (N):

$$S_{rate} = \left(\frac{\sum_{i=1}^{N} r_i}{N} \right) \times 100 \tag{6}$$

This metric distinguishes between consistent stability and intermittent success. For the primary surface analysis, $N \geq 20$; for the secondary wooden surface analysis, $N = 5$.

IV. Results

This section develops the quantitative data collected from the parameter sweep experiments. The results are grouped as controller tuning, trajectory performance with different velocities, and environmental sensitivity.

A. Proportional Gain Tuning

Initial experiments focused on isolating the optimal Proportional Gain (K_p). As it was hypothesized, the system showed clear modes of failure at extremes of the gain spectrum.

- **Under-damping** ($K_p < 0.4$)**:** The follower failed to produce enough angular velocity to match the leader's turn rate causing monotonic divergence until signal loss.

- **Over-damping** ($K_p > 1.8$)**:** The system exhibited critical instability. While the robot could track the turn, the steering corrections initiated high-frequency oscillations (visible shaking).
- **Optimal Gain** ($K_p = 0.8$)**:** This setting gave the best compromise, which minimises steady-state error without inducing sustained oscillation.

B. Trajectory and Velocity Analysis

The following sequence of graphs (Figs. 4 to 6) display the system's tracking performance over the three curvature profiles (Wide, Medium, Small) for various Base Speeds.

1) Performance at Low Velocity ($V_{base} = 26$): As shown in Fig. 4, lower speeds favoured the larger diameter paths. The Wide Circle (Blue) remained stable. However, the Small Circle (Grey) struggled to maintain the link, often lagging behind due to insufficient angular momentum to overcome turning friction.

Fig. 4. Tracking performance at Base Speed 26. The Wide Circle (Blue) tracks effectively, while the Small Circle (Grey) fails to complete the trajectory.

2) Performance at Intermediate Velocity ($V_{base} = 27$): Fig. 5 illustrates the transition phase. The stability of the Wide Circle begins to degrade, while the Small Circle (Green) shows signs of divergence, peaking at 4.5cm.

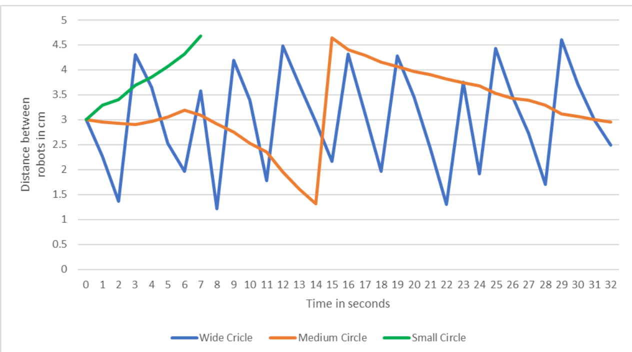

Fig. 5. Tracking performance at Base Speed 27. Note the increased amplitude of oscillation for the Wide Circle (Blue) compared to the lower speed test.

3) Performance at High Velocity ($V_{base} = 28$): Fig. 6 confirms the velocity-curvature trade-off. At higher speeds, the Wide Circle (Blue) consistently overshoots the trajectory (sawtooth divergence). Conversely, the **Small Circle (Grey)** achieves its optimal performance, maintaining a tight lock ($\approx$ 3cm). The increased velocity helps the differential drive overcome the static friction on the tight turn.

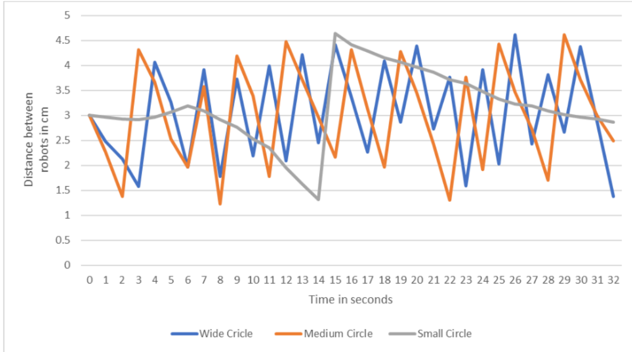

Fig. 6. Tracking performance at Base Speed 28. The Small Circle (Grey) exhibits the most stable tracking of the entire dataset, while the Wide Circle (Blue) diverges.

4) Combined optimal velocity for circles: Fig. 7 demonstrates the depiction of the experiments with the best value for different circles. Performance of the robot in a small circle (Grey) at $V_{base} = 28$, Performance of the robot in a medium circle (Orange) at $V_{base} = 27$, and Performance of the robot in a wide circle (Blue) at $V_{base} = 26$.

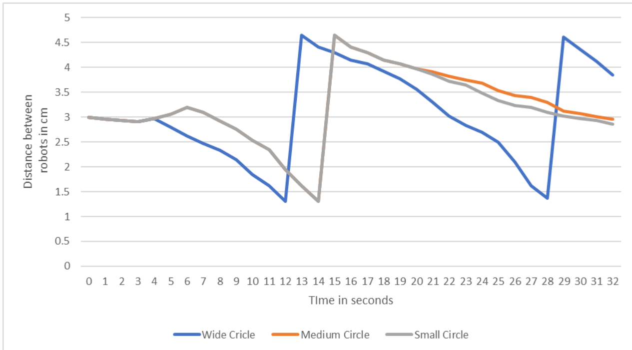

Fig. 7. Combined results of best performance at optimal Base Speed values. Small circle ($V_{base} = 28$), medium circle ($V_{base} = 27$), Wide circle ($V_{base} = 26$).

The graph shows closely overlapping lines for small and medium circles, implying that they had similar performance to a certain degree. However, the line for the wide circle shows irregularities, which can be addressed using a motor value with a decimal expansion to overlap with the other two lines.

C. Environmental Sensitivity

To quantify the impact of environmental factors, identical control codes ($V_{base} = 28, K_p = 0.8$) were executed on two distinct surfaces, as shown in Fig. 8.

- **Smooth Surface (Blue Line):** The robot maintains a stable following distance, oscillating between 2cm and 4.5cm. This bounded behavior indicates a stable control loop.
- **Rough Surface (Orange Line):** The system exhibits immediate divergence. The increased friction effectively dampens the steering response (under-steering). The distance increases linearly from 3cm to > 5.5cm within 16 seconds, resulting in total signal loss.

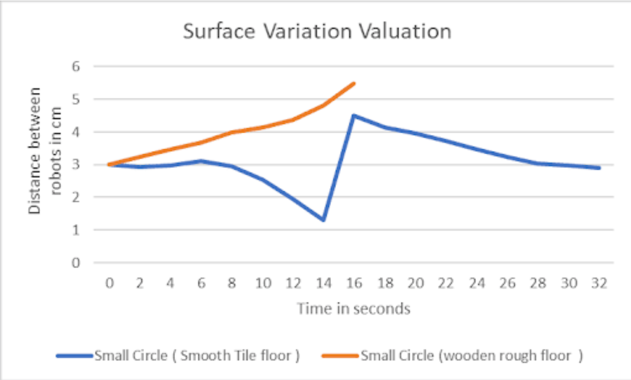

Fig. 8. Impact of surface texture on tracking stability. The Smooth Tile (Blue) maintains bounded oscillation, while the Rough Wood (Orange) causes linear divergence and eventual failure.

V. Discussion and Conclusion

A. Re-statement of Hypothesis

As proposed in Section III, this study was guided by the following hypothesis:

> *The tracking stability of the leader-follower system is governed by an inverse relationship between **Path Curvature** and **Maximum Stable Velocity**. We hypothesize that as the path curvature increases (tighter circles), the follower will require a proportionally higher control gain (K_p) to prevent under-steering. However, this increased gain will effectively lower the critical velocity threshold for oscillation, creating a narrowing "stability corridor."*

B. Discussion of Results

The experimental data provides strong support for the "Optimal Gain Range" hypothesis but presents a more complex reality regarding the "Velocity-Curvature" relationship.

1) The Gain Trade-off (Supported): The results confirmed that a fixed Proportional Gain represents a compromise. At $K_p < 0.4$, the system failed to track sharp turns due to under-damping, while values approaching $K_p \approx 2.0$ introduced critical instability. The experimentally derived optimal ($K_p = 0.8$) supports the prediction that a specific stability window exists.

2) The Velocity Paradox (Partially Refuted): The hypothesis predicted that tighter circles would strictly require *lower* speeds to maintain stability. However, the results (Fig. 6) revealed the opposite for the smallest radius ($D = 20.5$cm). The follower actually performed *better* at a higher base speed ($V = 28$) compared to the lower speed ($V = 26$).

Analysis: This abnormality suggests that for differential drive robots on high-friction surfaces, **static friction** dominates at low speeds. On the tightest turn, the inner wheels likely stalled or slipped at low velocities. Increasing the speed provided sufficient angular momentum to overcome this static friction, allowing for smoother tracking. This indicates that the "Stability Corridor" is bounded not just by sensor limits (high speed), but also by mechanical friction limits (low speed).

3) Environmental Sensitivity (Supported): The comparison between smooth and rough surfaces (Fig. 8) clearly supported the hypothesis. The loss of signal observed on the wooden surface demonstrates that the Proportional controller is not robust to environmental changes. The increased friction coefficient of the wood effectively "dampened" the robot's turning response, treating the optimized $K_p = 0.8$ as an under-damped setting.

C. Conclusion

This report investigated the feasibility of implementing a leader-follower system using low-cost IR sensing. The study concludes that while implicit optical communication is a viable low-latency solution, it is highly sensitive to physical dynamics.

The key finding is that **multi-robot coordination does not strictly require complex sensors** (like LIDAR), but it does require environment-specific tuning [2]. We demonstrated that a simple Proportional controller can achieve $\pm$2cm accuracy, provided the Base Speed is matched to the trajectory curvature.

D. Limitations and Future Work

The primary limitation of this study was the use of a fixed-parameter controller (Static K_p). As shown in the results, a gain that worked for the Medium circle failed for the Wide circle.

- **Future Work:** This could be rectified by implementing **Gain Scheduling** or an **Adaptive PID** controller. In such a system, the robot would dynamically increase K_p when the error is large (sharp turn) and decrease it when the error is small (straight line). Additionally, incorporating a longitudinal Time-of-Flight sensor would remove the need for manual speed calibration, allowing the follower to actively manage the following distance to stay within the optimal "linear region" of the IR sensor.

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